Maximum likelihood estimation (MLE) recap

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  - We may want to **infer** causality/associations (e.g., how lifestyle choices influence lifespan).

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- Probability distribution models are most commonly fit by **maximum likelihood estimation** (MLE).

Gaussian (normal) distribution

- Consider the Gaussian distribution, given by a density function

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

where $\mu \in \mathbb{R}, \sigma^2 \in \mathbb{R}_+$ are **parameters**.

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- We are given some data e.g., measurements of physical height. E.g. we may have $n = 5$ observations like this:

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- Our task is to estimate the parameters from the data.

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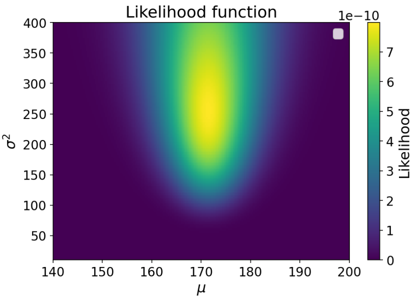
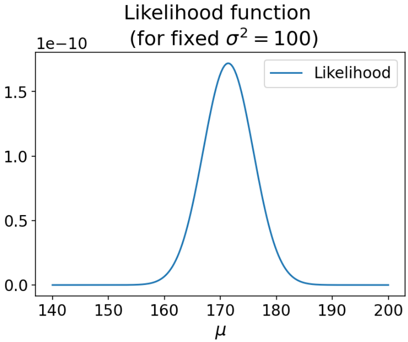
- We may plot the likelihood (for a specific data X from the example above):





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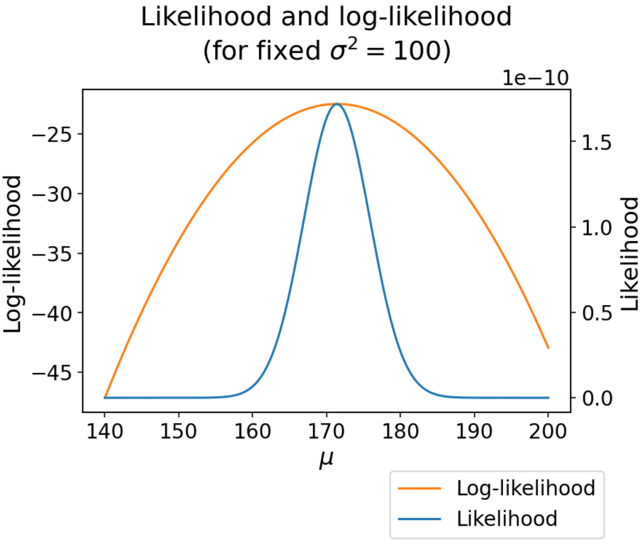
- So instead of maximizing likelihood, we may equivalently maximize log-likelihood, given as:

$$l(\mu, \sigma^2) = \log\left(L(\mu, \sigma^2)\right) = \log\left(\prod_{i=1}^{n} f(x_i \mid \mu, \sigma^2)\right)$$
$$= \sum_{i=1}^{n} \log f(x_i \mid \mu, \sigma^2)$$

Log-likelihood: visualization



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$$\begin{aligned} l(\mu, \sigma^2) &= \log\left(\prod_{i=1}^{n} f(x_i \mid \mu, \sigma^2)\right) \\ &= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2 \end{aligned}$$

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- By setting the derivative to 0 and solving for μ, we get

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$